

Deep Learning

Project

- Select 2 datasets
- Apply data preprocessing (scaling, augmentation, ...)
- Build DL model
- Train model & plot learning curves
- Test model & display evaluation metrics + FLOPS
- Classification: ROCAUC, ACC, F1, Kappa, Recall, Precision
- Regression: MAE, MAPE, RMSE, R^2
- Energy consumption, and carbon emission

Assignment 1:

- Vision Transformer
- Architecture
- Self-Attention: Math Example
- Applications

Assignment 2:

- Diffusion Model
- Architecture
- Math Formula
- Applications